

Fields 1

Arham Lodha

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Problem 1.

Proof. Let R be a ring.

($\implies$): Suppose $(0) \subset R$ is a maximal ideal. Then $\forall r \in R \ni r \neq 0, (r) = (1) = R$. By definition of ideal, $\exists a \in R \ni ar = 1$, r invertible. Thus R is a field.

($\impliedby$): Suppose R is a field. Let I be an ideal. If $\exists r \in I \ni r$ invertible, then $r^{-1}r = 1 \in I$, thus $I = (1) = R$. Otherwise, $I = (0)$ because 0 is the only noninvertible element. Thus (0) is maximal. $\square$

Problem 2.

Proof. $\forall p, q \in F[X] \ni p \neq 0$ and $q \neq 0$. $p(X) = \sum_{k=1}^{\deg p} a_k X^k$ and $q(X) = \sum_{k=1}^{\deg q} b_k X^k$. The highest degree term of $p(X)q(X)$ is $a_k b_k X^k$, $a_k b_k \neq 0$, since $a_k \neq 0$ and $b_k \neq 0$ and fields are integral domains. Thus $p(X)q(X) \neq 0$ and $F[X]$ is an integral domain.

Let $I \subset F[X]$ be an ideal. Let $f \in I$ be a nonzero element with minimal degree. $\forall p \in F[X]$, since $F[X]$ is a euclidean domain, $\exists q, r \in F[X] \ni p = fq + r$ where $\deg r < \deg f$. Thus $r = p - fq \implies r \in I$. Thus $r = 0$, otherwise contradiction since $\deg r < \deg f$ and f is the nonzero element with minimal degree in I . Thus $f|p$ and $I = (f)$.

Suppose $f_1, f_2 \in I, \ni I = (f_1) = (f_2) \implies \exists p, q \in F[X] \ni f_1 = pf_2, f_2 = qf_1$. Thus $f_1 = pqf_1 \implies f_1(1 - pq) = 0 \implies 1 - pq = 0 \implies pq = 1 \implies p, q \in F$. Thus f_1 and f_2 are scalar multiples of each other. Thus each ideal I exists a unique monic polynomial f such that $I = (f)$. $\square$

Problem 3.

Proof. Let $\pi : R \rightarrow R/I$. Let S be the set of ideals of R which contain I . Let S' be the set of ideals of R/I . There exists a bijection, $S \rightarrow S'$ such that $J \mapsto \pi_*(J)$ and an inverse map $\bar{J} \mapsto \pi^*(\bar{J})$.

$\forall J \in S, \pi^*(\pi_*(J)) = \pi^*(\{j + I | j \in J\}) = \{j | j \in J\} = J$. $\forall \bar{J} \in S', \pi_*(\pi^*(\bar{J})) = \pi_*(\{a | a \in j + I, j + I \in \bar{J}\}) = \{j + I, j + I \in \bar{J}\} = \bar{J}$. Thus π_* form a bijection.

Suppose $J_1 \subseteq J_2$ ideals of R containing I , then $\pi_*(J_1) = \{j_1 + I | j_1 \in J_1\} \subseteq \{j_2 + I | j_2 \in J_2\} = \pi_*(J_2)$, in R/I . This respects the inclusion relation, thus π_* is order preserving.

Let $\bar{R} := R/I$.

Suppose $J \subseteq R$ is prime and $I \subseteq J$. Then by definition, R/J is an integral domain. Let $\bar{J} = \pi_*(J)$.

$$\begin{array}{ccc}
R & \xrightarrow{\quad} & R/I \\
\downarrow & & \downarrow \\
R/J & \xrightarrow[\cong]{} & \overline{R}/\overline{J}
\end{array}$$

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The kernel of the map of $R \rightarrow \overline{R}/\overline{J}$ is J . Thus $\overline{R}/\overline{J} \cong R/J$. Thus, since R/J is a integral domain, $\overline{R}/\overline{J}$, thus $\overline{J}$ is prime. $\square$

Problem 4.

Proof. $\implies$: Suppose $P \subset R$ is a prime ideal. Then R/P is a integral domain, thus $\forall a + P, b + P \in R/P \ni (a + P)(b + P) = 0 \iff a + P = 0$ or $b + P = 0$. If $a + P = 0 \implies a \in P$. If $b + P = 0 \implies b \in P$.

$\Leftarrow$: Suppose P is an ideal with the following property $\forall a, b \in R \ni ab \in P \implies a \in P$ or $b \in P$, then for $(a + P), (b + P) \in R/P$, $(a + P)(b + P) = P = 0 \implies a + P = P = 0$ or $b + P = P = 0$. Thus R/P is a integral domain. Thus P is a prime ideal. $\square$

Problem 5a. Show that in a PID a proper ideal is prime iff it is 0, or maximal.

Proof. Let R be a PID. ($\implies$) Let $I = (r)$ be a prime ideal of R . It suffices to show that if $r \neq 0 \implies I$ is maximal. Let $m \in R \ni (r) \subset (m) \implies \exists n \in R \ni r = mn \implies mn \in (r)$. By definition of prime ideal, $m \in I$ or $n \in I$. If $m \in I \implies (m) \subseteq (r) \implies (m) = I$. If $n \in I \implies \exists p \in R \ni n = pr \implies r = mpr \implies r(1 - mp) = 0$. Since R is a PID, $1 - mp = 0 \implies mp = 1$ thus m unit and $(m) = R$.

$\Leftarrow$: Suppose $I = (r)$ where $r \neq 1$ and $r = 0$ or I maximal. If $r = 0$, then $\forall a, b \in R \ni ab \in I \implies ab = 0$, since R is PID, $a = 0$ or $b = 0$. Thus (0) is prime. Suppose I maximal and $r \neq 0$, then R/I is a field and thus a PID, thus I is prime. $\square$

Problem 5b. Show that for any non-zero prime ideal, the quotient is a field

Proof. For every non zero prime ideal $I = (r)$, (r) is maximal, thus the only maximal ideal of $R/(r)$ is (0) , thus by 1 $R/(r)$ is a field. $\square$

Problem 6.

Proof. $I = (f(x))$. Let $\phi : F^{\deg f} \rightarrow F[X]/(f(x))$ where $(f_0, \dots, f_{\deg f - 1}) \mapsto \sum_{i=0}^{\deg f - 1} f_i x^i + I$. $\forall (f_0, \dots, f_{\deg f - 1}), (g_0, \dots, g_{\deg f - 1}) \in F^{\deg f}$ and $a, b \in F$, $\phi(a(f_0, \dots, f_{\deg f - 1}) + b(g_0, \dots, g_{\deg f - 1})) = (\sum_{i=0}^{\deg f - 1} (af_i + bg_i)x^i) + I = a\phi((f_0, \dots, f_{\deg f - 1})) + b\phi((g_0, \dots, g_{\deg f - 1}))$. Thus ϕ is a linear map.

$\dim F[X]/(f(x)) = \deg f = F^{\deg f}$, because $F[X]/(f(x))$ is the set of all polynomials up to degree $\deg f$.

$\forall k \in \ker \phi$, $k = (0, \dots, 0)$, because $\phi(k) = 0 \implies k_i = 0$.

Thus ϕ is a isomorphism. Thus $F^{\deg f} \cong F[X]/(f(x))$ $\square$

Problem 6a.

Proof. Since $f(x)$ is a nonconstant $\implies (x)$ is nonzero.

$(a \implies b)$: S is a integral domain $\iff (f)$ is prime $\iff S$ is a field, by 5.

$(b \implies c)$: If S is a field, then it is a PID, $\implies (f)$ is prime. Since (f) is prime $\iff (f)$ is maximal. $\forall a, b \in F[X] \ni f = ab \implies f \in (a) \implies (f) \subseteq (a)$. If inclusion was proper, $(a) = F[X] \implies a \in F$ and thus unit. If inclusion is not proper and $(f) = (a)$, by 1, a is a simple scalar multiple of f , and thus b is the scalar. By symmetry, b is a scalar or a scalar multiple of f (a is the scalar). Thus f is irreducible.

$(c \implies a)$: If f is irreducible, then $\forall a, b \in F[X] \ni f = ab \implies a$ or b are scalars. f is only divisible by a scalar or itself. Thus (f) is prime, thus S is a integral domain.

Thus $a \implies b \implies c \implies a$, and $a \iff b$ and $b \iff c$ and $c \iff a$. $\square$

Problem 7.

Proof. $S = F_2[X]/(x^3 + x + 1) \cong F_2 \oplus F_2x \oplus F_2x^2$, thus $|S| = 8$. Similarly $R = F_2[X]/(x^3 + x^2 + 1) \cong F_2 \oplus F_2x \oplus F_2x^2$, thus $|R| = 8$. Since $x^3 + x + 1$ irreducible, S is a field. Similarly since $x^3 + x^2 + 1$ is irreducible, R is a field.

$$\begin{array}{ccc} & & F_2[x] \\ & \swarrow \alpha: [x \mapsto x+1] & \downarrow \beta \\ F_2[x]/(x^3 + x^2 + 1) & \xleftarrow{\quad} & F_2[x]/(x^3 + x + 1) \end{array}$$

α sends $F_2 \rightarrow F_2$ and $x \rightarrow x + 1$. But $(x + 1)^3 + (x + 1)^2 + 1 = x^3 + x + 1$. Thus α and β have the same kernel. Thus there exists a homomorphism from $R \rightarrow S$. Since there exists a map it is injective. Since $|S| = |R|$, the map is surjective and thus a isomorphism. $\square$

Problem 8.

Proof. $(a \implies b)$: $F[s]$ being a finite dimensional F vectorspace. $\forall t \in F[s] \ni t \neq 0$. Let $\phi: F[s] \rightarrow F[s]$ where $r \mapsto tr$. $\forall k \in \ker \phi$, $\phi(k) = tk = 0$, since $F[s]$ is a PID, $k = 0$. Thus ϕ is injective. $\dim F[s] = \dim F[s]$ so ϕ is also surjective. Thus there is a bijection. Thus $\exists r \in F[s] \ni \phi(r) = 1 \implies tr = 1 \implies r = t^{-1}$. Thus $F[s]$ is a field.

$(b \implies c)$: Suppose $F[s]$ is a field. Take the ring homomorphism, $\phi: F[X] \rightarrow F[s]$ where $f \mapsto f(s)$. ϕ is surjective, because every element of $F[s]$ is s evaluated on a polynomial with coefficients of F . Assume the contrary, $\ker \phi$ is trivial, then ϕ is an isomorphism and $F[X] \cong F[s]$ thus $F[X]$ is a field which is a contradiction. Thus $\ker \phi$ is nontrivial. Since $F[X]$ is a PID, $\exists f \in F[X] \ni \ker \phi = (f)$. Thus $\phi(f) = f(s) = 0$.

$(c \implies a)$: Suppose $\exists f \in F[X] \ni f(s) = 0$. Let $n := \deg f$. Thus there is a relation that $\sum_{k=0}^n a_k s^k = 0 \implies s^n = \sum_{k=0}^{n-1} -\frac{a_k}{a_n} s^k$. Let $B = \{1, s, \dots, s^{n-1}\}$

Thus $\forall t \in F[s], t = \sum_{k=0}^m c_k s^k$, if $m < n$, t is a linear combination of B .

If $m = n$, $t = \sum_{k=0}^m c_k s^k = \sum_{k=0}^{n-1} c_k s^k + c_n \sum_{k=0}^{n-1} -\frac{a_k}{a_n} s^k \implies t$ is a linear combination of B .

If $n < m$, then $t = \sum_{k=0}^{n-1} c_k s^k + \sum_{k=n}^m c_k s^k = \sum_{k=0}^{n-1} c_k s^k + \sum_{k=n}^m c_k s^{k-n} s^n = \sum_{k=0}^{n-1} c_k s^k + \sum_{k=n}^m s^k (\sum_{l=0}^{n-1} -\frac{a_l}{a_n} s^l)$. Evaluate t , and then replug, $\sum_{k=0}^{n-1} -\frac{a_k}{a_n} s^k$ into s^n , and then reevaluate and

for every $s^k \ni k > n$ write s^k as $s^k = s^{k-n}s^n$. Repeat this process, till no s^n and larger powers of s remain. Thus t can be written as a linear combination of B .

Thus every element of $F[s]$ can be written as a linear combination of B . Thus $F[s]$ is a finite dimensional F -vector space.

Thus $a \implies b \implies c \implies a$, meaning $a \iff b$ and $b \iff c$ and $c \iff a$, because the chain of implications are cyclic. $\square$

Problem 8b. Suppose s is not algebraic, show $F[s]$ is isomorphic to $F[X]$.

Proof. Let $\phi : F[X] \rightarrow F[s]$ where $f \rightarrow f(s)$. ϕ is surjective, because $\forall t \in F[s]$, $t = \sum_{k=1}^n a_k s^k$ let $\phi(\sum_{k=1}^n a_k X^k) = t$. Since s is not algebraic, by definition $\nexists f \in F[X] \ni f(s) = 0$. Thus $\ker \phi = 0$. Thus ϕ is an isomorphism.

Thus $F[s] \cong F[s]$ $\square$

Problem 9.

Proof. Let M be a S -Module. Define an action of R on M where $\forall r \in R$ and $\forall m \in M$, $r \cdot m = f(r)m$. Since f is a Ring homomorphism the action is linear. Thus M is a R -Module.

$V = IM$ is closed under addition because of the way IM is defined as the all possible sums of all elements of the form im where $i \in I$ and $m \in M$. $\forall n \in V$, $n = im$, for some $i \in I$ and $m \in M$ and $\forall r \in R$, $rv = rim$, $ri \in I$ (def of ideal), thus $rv \in V$.

There exists a linear action of R/I on M/MI where $\forall x + I \in R/I$ and $\forall m + MI \in M/MI$, $(x + I) * (m + MI) := xm + MI$. $\forall i \in I$, $((x + i) + I) * (m + MI) = (x + i)m + MI = xm + im + MI = xm + MI$. The action is well defined. Thus M/IM is a R/I module, and thus a restriction of a R/I module. Suppose N is a restriction of a R/I module. Let $f : M \rightarrow N$ be any M linear homomorphism between the modules. f must send elements of IM to 0, thus $IM \leq \ker f$. Thus f must factor through the quotient map $M \rightarrow M/IM$. Since the quotient map is surjective, it must factor through uniquely.

Let $\alpha : M \rightarrow N$. Let π_N and π_M are the standard quotient maps. $f := \alpha \circ \pi_N$. $\ker \pi_N = IN$. $\alpha^*(\ker \pi_N) = \alpha^*(IN) = IM$. Thus $IM \leq \ker f$. f must factor through π_M and thus $\exists \beta : M/IM \rightarrow N/IN$ and $f = \beta \circ \pi_M$. If the f is surjective, since π_N is surjective, β must also be surjective.

$$\begin{array}{ccc}
 & N & \xrightarrow{\pi_N} N/IN \\
 \alpha \nearrow & & \searrow f \\
 M & \xrightarrow{\pi_M} M/IM & \searrow \beta
 \end{array}$$

Let $\varphi : R^k \rightarrow (R/I)^k$ where $(r_1, \dots, r_k) \rightarrow (r_1 + I, \dots, r_k + I)$. $\forall a, b \in R^k$ and $\varphi(a(r_1, \dots, r_k) + b(s_1, \dots, s_k)) = (ar_1 + bs_2 + I, \dots, ar_k + bs_k + I) = a(r_1 + I, \dots, r_k + I) + b(s_1 + I, \dots, s_k + I) = a\varphi((r_1, \dots, r_k)) + b\varphi((s_1, \dots, s_k))$, φ is a homomorphism. $\ker \varphi = IR^k$. By the first isomorphism theorem of modules, $R^k/IR^k \cong (R/I)^k$.

Suppose we have $R^k \twoheadrightarrow R^j$. Let I be a maximal ideal. Thus R/I is a vectorspace and thus $R^k/IR^k \cong (R/I)^k$ similarly, $R^j/IR^j \cong (R/I)^j$ which are vectorspaces. Thus since $R^k \twoheadrightarrow R^j$ is surjective, $\exists$ a surjective map from $(R/I)^k \twoheadrightarrow (R/I)^j$, which is a map of modules, but R/I is a field, the map is a map of vectorspaces. Thus $j \leq k$. $\square$